VIRTUAL SCREENING II

Dr. Oluyemi Wande | Dr. Adeniyi Adewumi | Shadrach Eze

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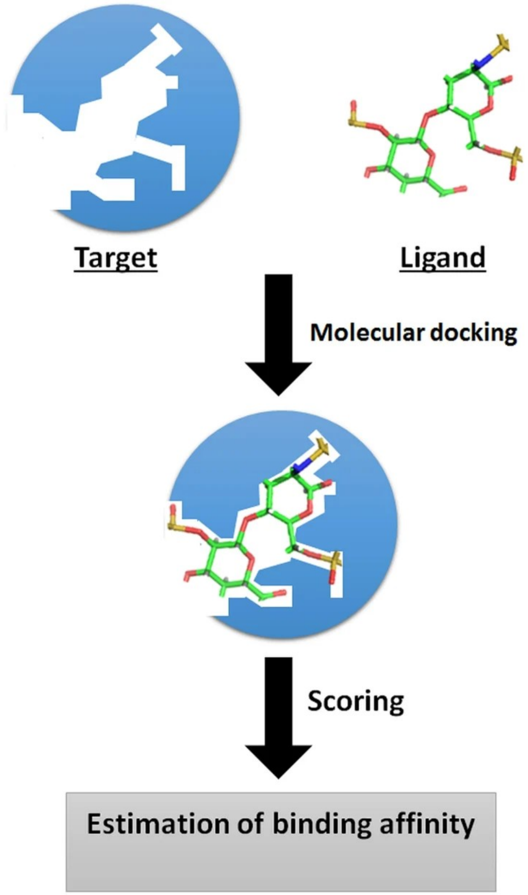
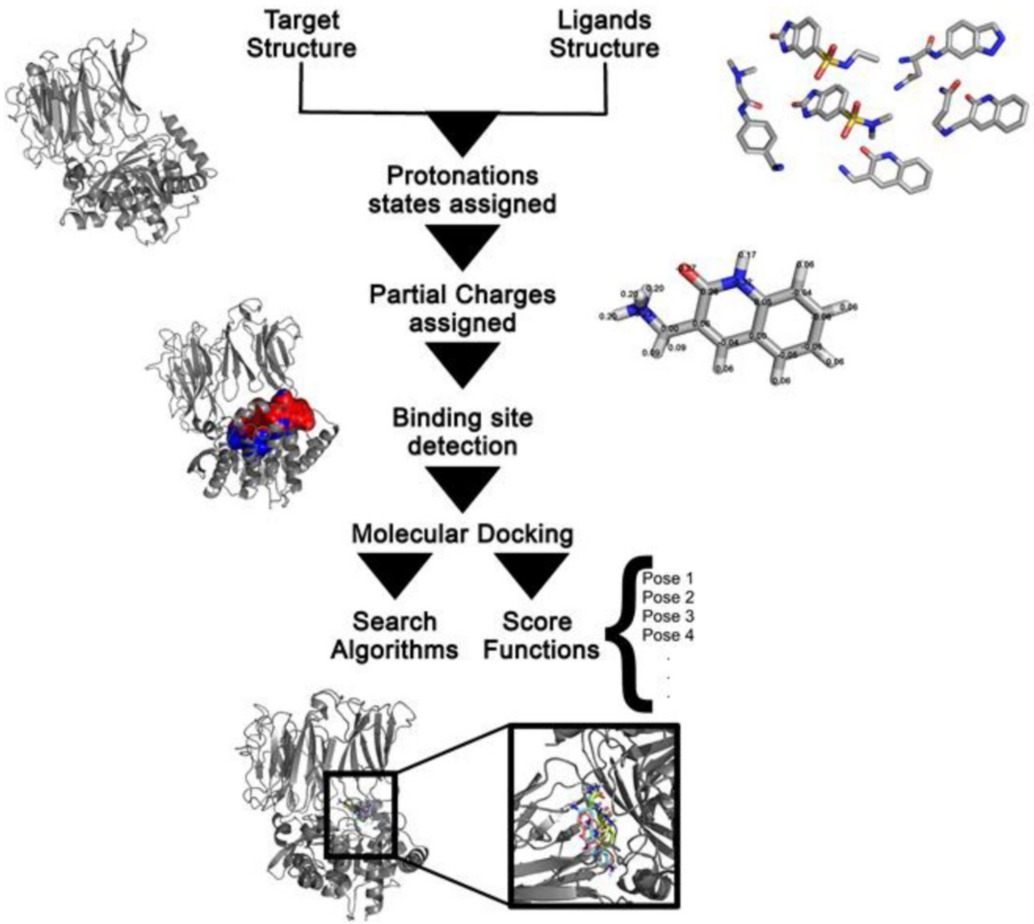
WHAT IS DOCKING?



Concept



Workflow

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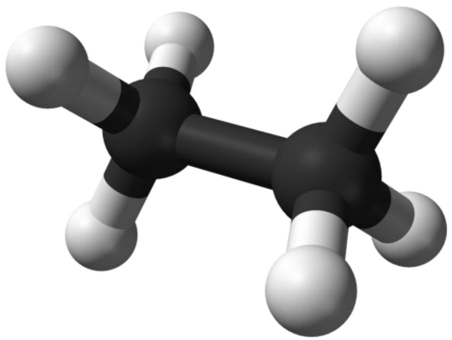
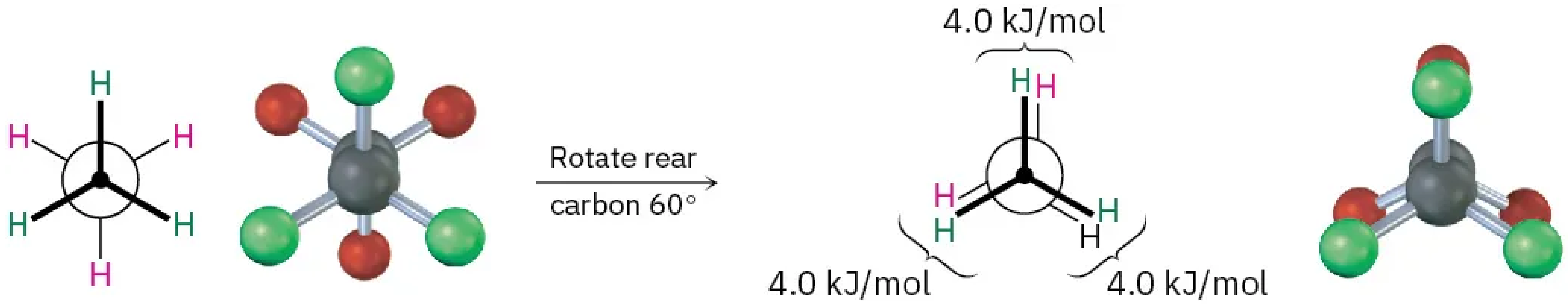
POSE (CONFORMATION)

What is the Pose or Conformation of a molecule?





Ethane—staggered conformation

Ethane—eclipsed conformation

POSE SEARCH

How do I explore all the possible poses that the molecule can adopt in the presence of the protein?

- **Search all the conformational space of the molecule and**
- **Possibly side chains of the protein in the active site**

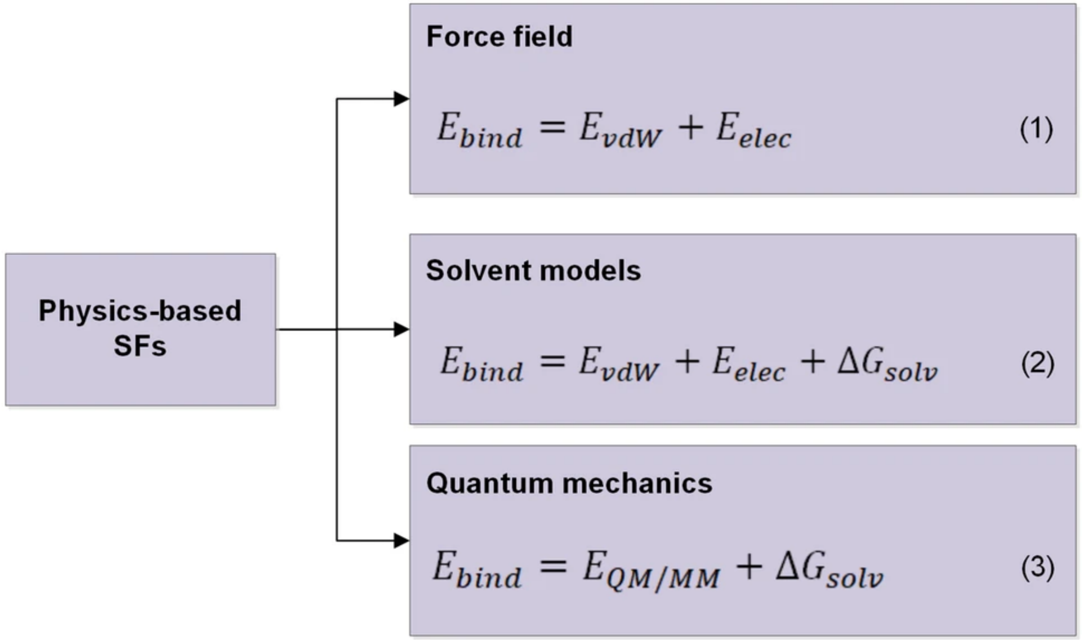
- **Translate the molecule (in X, Y, Z, direction) or**
- **Rotate the molecule along torsional angles**

- **Other faster variations exist**
- **Stochastic search**
- **Monte Carlo search**
- **Genetic Algorithm**

SCORING

Now that I have a pose, how do I assign a number that represents the correctness of the pose?

- We can consider all the van der Waals interactions, electrostatic interactions, hydrogen bonding etc
- Solvation effects and quantum mechanics refinements can also be added
- This is summed into a number as the **binding score**



Li, J., Fu, A. & Zhang, L. (2019). https://doi.org/10.1007/s12539-019-00327-w

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SCORING

Scoring Functions

Physics-based

DOCK: $$E_{bind} = \sum_{i=1}^{L}\sum_{j=1}^{R}\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}} + \frac{q_i q_j}{\varepsilon(r_{ij})r_{ij}}\right) \quad (1)$$

Empirical

X-Score: $$E_{bind} = w_0 + w_1\Delta G_{vdW} + w_2\Delta G_{Hbond} + w_3\Delta G_{rot} + w_4\Delta G_{hydro} \quad (2)$$

Knowledge-based

PMF: $$E_{bind} = \sum_{i=1}^{L}\sum_{j=1}^{R} -k_B T \ln[g(r)] \quad (3)$$

Machine learning-based

RF-Score: $$E_{bind} = f_{RF}(x_m) \quad (4)$$
$$x_{ij} = \sum_{i=1}^{L}\sum_{j=1}^{R}\Theta[d_{cutoff} - d_{ij}]$$

Li, J., Fu, A. & Zhang, L. (2019). https://doi.org/10.1007/s12539-019-00327-w

Force-Field-Based	Empirical	Knowledge-Based
DOCK [32]	AutoDock [34]	SMoG [82]
AutoDock [34]	GlideScore [37]	DrugScore [62]
GoldScore [35]	ChemScore [60]	PMF_Score [83]
ICM [46]	X_Score [66]	MotifScore [84]
LigandFit [47]	F_Score [73]	RF_Score [85]
Molegro Virtual Docker [48]	Fresno [75]	PESD_SVM [86]
SYBYL_G-Score [73]	SCORE [76]	PoseScore [87]
SYBYL_D-Score [73]	LUDI [77]	
MedusaScore [74]	SFCscore [78]	
	HYDE [79]	
	LigScore [80]	
	PLP [81]	

Many docking algorithm uses one or more of these methods

Ferreira, et. al. (2015) https://doi.org/10.3390/molecules200713384

RANKING

Now that my poses have scores, which of them should come first or last in an ordered list

Several Factors

- Best Docking Score
- Pose with lowest RMSD
- RMSD of pose with the best docking score
- docking score of pose with lowest RMSD

- **Ranking can be done based on several factors depending on the goal of the docking campaign**
- **Inter-criteria Analysis can be used here**

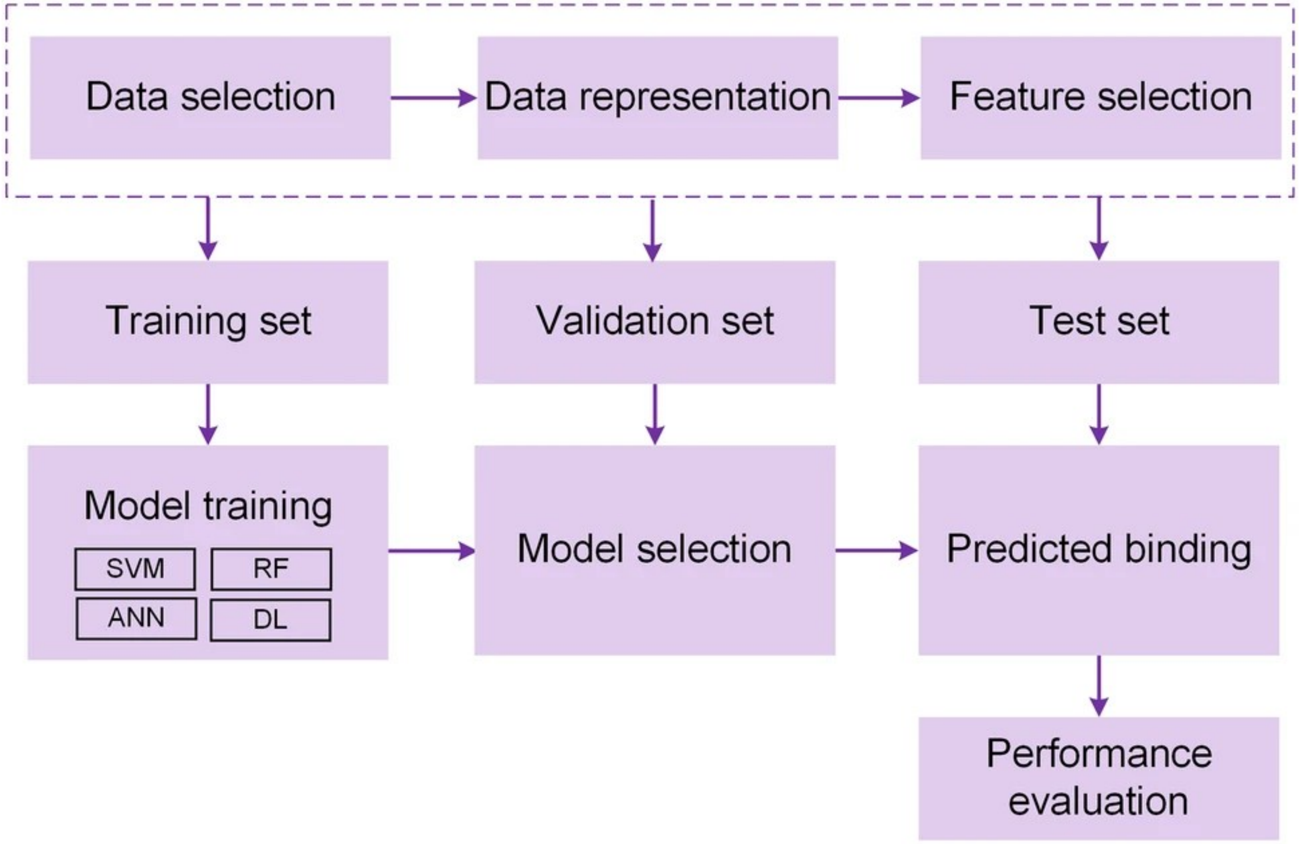
https://doi.org/10.3390/molecules30132777

ALGORITHMS

Algorithm	Milestone	Reference
GRAMM	Rigid docking, six-dimensional shape complementarity; fast Fourier transformation	[55]
FTDock	Implementation of electrostatics and biochemical information	[56,57]
3D-Dock	Additionally, energy calculations, side chain optimization, and backbone refinement	[58]
Hex	Spherical polar Fourier correlation method	[59]
Dot/Dot2	Implementation of Poisson–Boltzmann methods	[60,61]
HADDOCK	Flexibility of amino acid side chains	[62]
PatchDock	Local feature matching instead of six-dimensional transformation fitting	[63]
ParaDock	Shape complementarity but flexible NA structure prediction	[64]
NPDock	Rigid body docking while considering the specific features of NA	[53]
HDOCK	Docking between two big molecules; template-based and template-free rigid docking mode	[65,66]
Gold	Full flexibility or rotamer-based search for both ligand and selected amino acids residues; docking in a determined binding pocket. Presents a range of different scoring functions, from machine-learning-based to physicochemical-based ones	[67]
Autodock Autodock Vina	Full flexibility or rotamer-based search for both ligand and selected amino acids residues; docking in a determined binding pocket. Energy-based scoring function and ability to handle surface pockets	[68]

Krüger et. al. (2018) https://doi.org/10.3390/biom8030083

ML-BASED SCORING FUNCTION



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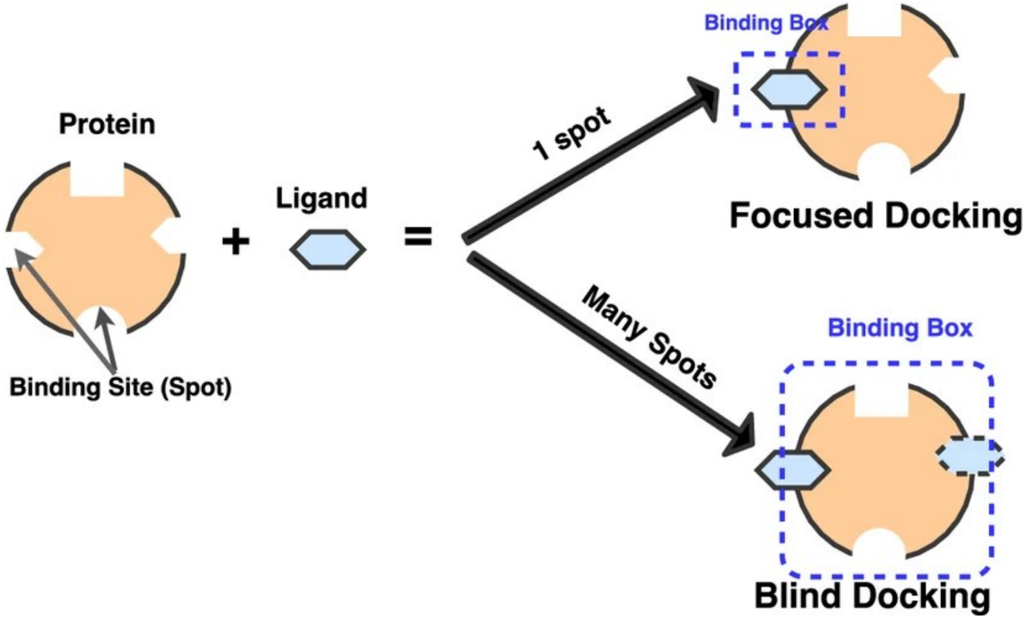
Also referred to as Machine Learning Interaction Potentials (MLiP)

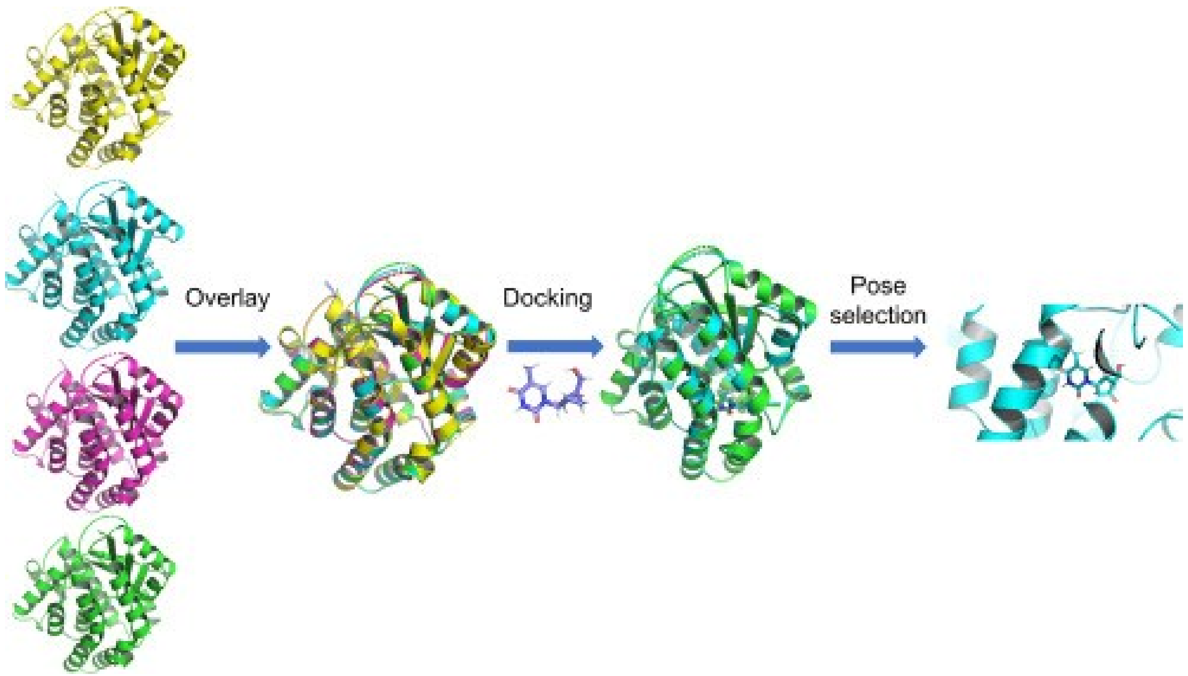
Here, we use neural networks and embedding to build or refine the pose selection and scoring

DOCKING VARIATIONS

- Blind Docking
  - Targeted Docking
- Covalent Docking
- Reverse Docking
- Flexible Docking
  - Rigid Docking
- Ensemble Docking

DOCKING VARIATIONS





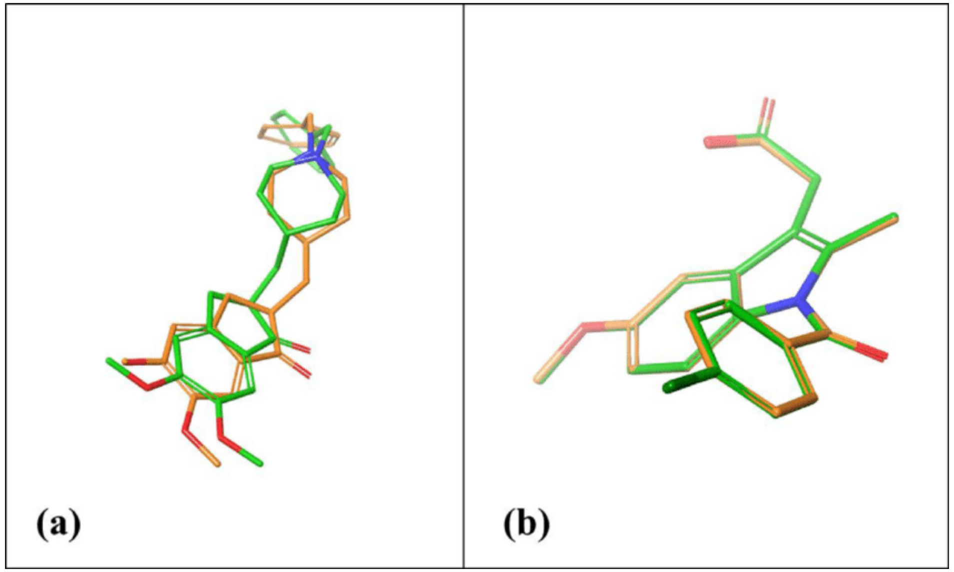
Blind vs Targeted Docking

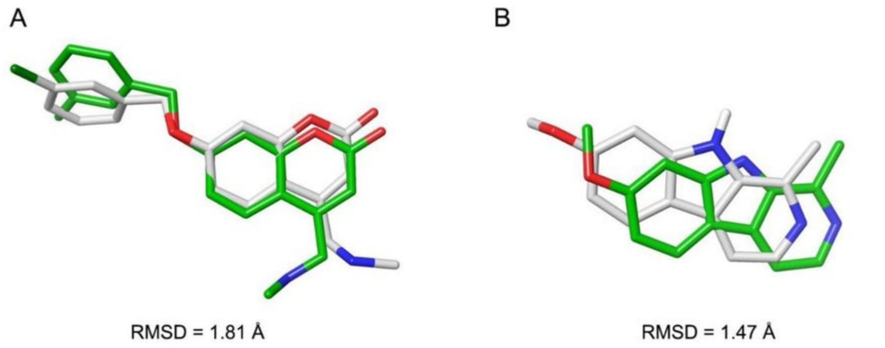
Ensemble Docking

DOCKING VALIDATION

Pose Reproduction

- How well does your docking pose deviate from known poses from crystallographic structures
- Important concept in validating a new docking algorithm





- Docking and redocking
- Docked pose vs crystal pose tracked by the RMSD
- Generally, an RMSD of < 2.0A is acceptable as very similar poses
- The lower the RMSD, the better

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DOCKING VALIDATION

Enrichment Factor

How much better is your docking protocol at finding "needles in a haystack" compared to a random selection

$$EF = \frac{a/n}{A/N}$$

a = Number of active compounds in the top **n** ranked compounds

n = Number of compounds selected from the top of the ranked list

A = Total number of active compounds in the entire database

N = Total number of compounds in the entire database

- The higher the EF, the better our protocol
- Gives us confidence on how our method can detect active compounds from Inactive compounds
- This is usually good when benchmarked on known inactive compounds like the Directory of Useful Decoys (DUD) dataset

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SCORING FUNCTION VALIDATION

METRICS

Scoring power

Ability to reproduce experimental binding energy

Ranking power

Rank ligands for set of known proteins

Docking power/Pose reproduction

Screening power

Same as Enrichment Factor Above

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FURTHER REFERENCES

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